



Convergence of LRRK2 and GBA1 in the pathogenesis of PD

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Summary

Genome-wide association studies identified GBA1 and LRRK2 as most common risk factors for Parkinson's disease (PD). Previously Ysselstein D et al reported that LRRK2 kinase activity negatively regulates Glucocerebrosidase (GCase) activity in human dopaminergic neurons, suggesting a convergent mechanism underlying mutant LRRK2 and GBA1 in the pathogenesis of PD¹. In the current study, we aimed to investigate the mechanism by which LRRK2 regulates GCase activity. We utilized human dopaminergic neurons differentiated from LRRK2 patient and isogenic control iPSCs to conduct the study. We found that LRRK2 G2019S PD neurons showed increased phosphorylated rab10 levels, indicating a hyperactivation of LRRK2. Inhibition of LRRK2 kinase activity by MLI-2 treatment increased lysosomal GCase activity and attenuated the accumulation of oxidized dopamine in G2019S PD neurons. The levels of GCase, Saposin C, Cathepsin D and Cathepsin B were not changed in G2019S PD neurons, which suggested that mutant LRRK2 did not affect GCase expression levels or its activation. We further found that G2019S mutation did not affect post-ER GCase levels in an Endo H treatment assay. Additionally, the protein levels of GCase transporter LIMP2 were not changed. Together, these data raised the possibility that LRRK2 may regulate GCase activity in an indirect manner. Interesting, lysosomal membrane proteins LAMP1 and LAMP2 were increased in G2019S PD dopaminergic neurons. Further examination of lysosomal morphology and functionality in G2019S PD neurons will help define the role of LRRK2 in regulating GCase activity.

Method

PPMI lines

The information of the PPMI iPSC lines used in this study is as follows.

Specimen Barcode	PPMI ID	Cell Line Name (Clone ID)	Cell Type	Karyotype
CDI00008707	51518	11448.102	iPSC	46 XX





CDI00006063	56761	11325.103	iPSC	46 XY
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The G2019S mutation in PPMI iPSC lines 51518 and 56761 was successfully corrected, and two independent isogenic control lines were generated for each PD line. Dopaminergic neurons derived from iPSC line 51518 showed hyperactivation of LRRK2 kinase compared to isogenic control neurons, whereas DA neurons derived from 56761 line did not. Thus, only 51518 line was used for the rest of the study.

CRISPR/Cas9

SgRNA targeting sequence was cloned into Px458 backbone (Addgene #48138), and homology-directed repair (HDR) template (ssODN) was ordered from IDT. A silent mutation was introduced into the ssODN to avoid re-cutting. The DNA constructs were transfected into iPSCs by electroporation using Neon Transfection System (Thermo Fisher Scientific). Transfected cells were enriched by FAC-sorting of GFP-positive cells at 48 hours after transfection. After FAC-sorting, GFP+ cells were plated and cultured at low density on cultrex-coated 6-well plate. Single cell-derived colonies were picked and screened for positive clones. The correction of the mutation was validated by sanger sequencing. The constructs for G2019S correction and primers for genotyping are as follows.

G2019S sgRNA targeting site	GTCAGCAATCTTTGCAATGA
Backbone	Px458
G2019S-ssODN-sense (synonymous mutation is highlighted in red)	ACCGAGACCTGAAACCCCAACAATGTGCTGCTTTTCA CACTGTATCCCAATGCTGCgATCATTGCAAAGATTGC TGACTACGGCATTGCTCAGTACTGCTGTAGAATGGG GATAAAAACATCAGAGGGCACACCAGGTAGGTGAT CAGGTCTGTCTCATAATTCTATCTTCAGGATGGAT
Distance from cut site to mutation site	20bp

LRRK2 ^{G2019S} genotyping primers	G2019S-For	AGGGACAAAGTGAGCACAGA
	G2019S-Rev	CACTGGCCCCATAACTGCTT

Differentiation of midbrain dopaminergic neurons

Human iPSCs were differentiated into midbrain dopaminergic neurons as described previously². On day 13 of differentiation, cells were chunked (size of 1–2 mm) and replated onto poly-D-lysine (PDL)/laminin-coated 10 cm dish. At day 22–25, differentiated cells were dissociated and replated onto PDL/laminin coated plates for downstream assessments. Differentiated neurons were cultured using Neurobasal media (Life Technologies) supplemented with Neurocult SM1





(Stemcell technologies). Neuralization growth factors were added for the maturation of differentiated neurons.

Immunocytochemistry

Differentiated neurons cultured on coverslips were washed with PBS and fixed using 4% PFA. After fixation, cells were washed three times with PBS, then permeabilized with 0.1% saponin (with 10% FBS in PBS) for 1 hour at room temperature. Permeabilized cells were incubated with primary antibodies overnight at 4 °C. After incubation, cells were washed three times and incubated with anti-mouse or anti-rabbit secondary antibodies conjugated with AlexaFluor for 1h at room temperature protected from light. After washing, samples were mounted with DAPI+ mounting media and imaged on a confocal microscope. The following antibodies were used.

Target	Vendor	Catalog
TUJ1	BioLegend	801202
Tyrosine hydroxylase	Sigma	AB152

Immunoblotting

Neurons were collected at day 90 and lysed using RIPA lysis buffer with protease and phosphatase inhibitor cocktail (Thermo Scientific 78440). Samples were centrifuged at 16, 000x g for 15min at 4 °C, and the supernatant was collected. Protein concentration was measured using Pierce BCA Protein Assay Kit (Thermo Scientific 23225). Proteins were separated using Novex™ 4-20% Tris-Glycine protein gel, and transferred onto nitrocellulose membrane using BIO-RAD Trans-Blot Turbo Transfer system. Membranes were blocked with 5% nonfat milk and then incubated with primary antibody dilution overnight on an orbital shaker at 4 °C. Membranes were washed and then incubated with anti-mouse or anti-rabbit secondary antibodies conjugated with HRP. The signal was developed using BIO-RAD ChemiDoc system. The primary antibodies used in this study are as follows.

Target	Vendor	Catalog
Rab10	CST	8127P
P rab10 (T73)	Abcam	Ab230261
CTSD	Santa Cruz	sc-6487
CTSB	R&D system	AF953
Saposin C	Santa Cruz	sc-374119
GCase	Sigma	G4171
LAMP1	DSHB	H4A3
LAMP2	DSHB	H4B4
Tyrosine	Sigma	AB152





hydroxylase		
TUJ1	BioLegend	801202
GAPDH	millipore	MAB374

MLi-2 treatment

From day 80 of differentiation, LRRK2 G2019S PD dopaminergic neurons were treated with LRRK2 kinase inhibitor MLi-2, or DMSO as a negative control, every 2 days for 12 days prior to analysis, and the last treatment was performed 2 hours before harvesting the samples or conducting the live cell GCase activity assay.

Lysosomal GCase activity

Lysosomal GCase activity assay was conducted as previously described¹. LRRK2 G2019S PD iPSC-derived dopaminergic neurons were plated on 96-well plate (with glass bottom and black wall) coated with poly-D-lysine and laminin. Neurons were treated with DMSO or 600 nM MLi-2 from day 80 for 12 days before the assay. The neurons were washed with phenol red-free neurobasal media with SM1 supplement, and incubated with phenol red free media containing 50 µg/mL 5-(pentafluorobenzoylamino) fluorescein di-β-D-glucopyranoside (PFB-FDGlu, a specific GCase substrate) (Thermo Fisher) for 1 hour at 37 °C in dark to allow the substrate to accumulate in the lysosome. After incubation, the neurons were washed to remove substrate and then incubated with phenol red free media for 30 min at 37 °C in dark. GCase hydrolysis activity was measured as the unquenching of the fluorescent probe at 485/525 nm (excitation/emission) using a plate reader. After readying, the cells were lysed using RIPA lysis buffer and the proteins were quantified by BCA assay. The GCase activity was normalized to total protein for each well.

Detection of oxidized dopamine

Oxidized dopamine was quantified by near infrared fluorescence (nIRF) as previously described³. Neurons were washed and collected with ice cold PBS. After centrifugation at 200 g for 3 min, PBS was removed, and the cell pellet was resuspended in RIPA lysis buffer. The samples were homogenized and then lysed for 30min on ice. After centrifugation at 16, 000x g for 15min at 4 °C, the supernatant RIPA lysate was transferred to a new tube and the pellet was further lysed using 2% SDS lysis buffer (in 50mM Tris-HCl solution, pH 7.4) by boiling (at 100 °C for 10 min), sonification and boiling (at 100 °C for 10 min). Protein amount of the RIPA lysate was calculated and the volume of SDS lysis buffer was normalized to protein amount (20ul of SDS lysis buffer per 100 ug of protein) for each sample. Samples were centrifuged at 150, 000x g for 30 min at 4 °C. The supernatant was discarded, and the pellet was dissolved in 1N NaOH solution at 55 °C overnight. The samples were dried using speedvac and washed with nanopure water and then lyophilized again. The final pellet was dissolved using nanopure water. To make the oxidized dopamine standards, 10mM of dopamine was oxidized using 20mM NaIO₄ (in D-PBS), and a serial of standards were diluted from the stock solution using nanopure water. 10µl of standard or sample was dropped onto a Biodyne nylon transfer membrane (Pall,





#Pall-60209), and the membrane was air dried, and the signal was scanned using the 700 channel of an Odyssey infrared imaging system (Li-Cor). The intensity of each sample and standard were quantified using Image Studio.

Endo H digestion

Post-ER fraction of GCase proteins was quantified by Endo H treatment, as N-linked glycans of GCase proteins are converted in the Golgi to complex oligosaccharides that are resistant to Endo H digestion. On day 90 of differentiation, neurons were collected and lysed using RIPA lysis buffer. The treatments with Endo H or PNGase F were performed following the manufacturer's instruction. For each condition, 20 ug of proteins were denatured at 100 °C for 10min. After being equilibrated to room temperature, the samples were digested by either Endo H or PNGase F for 1 hour at 37 °C. The samples were boiled with laemmli SDS sample buffer and analyzed by western blot. The post ER and deglycosylated forms of GCase proteins were indicated as Endo H and PNGase F resistant fractions, respectively.

References

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